

PH 606: Quantum Field Theory

Course Code	PH 606
Course Name	Quantum Field Theory
L-T-P-C	3-0-0-3
Prerequisites	Quantum mechanics (PH 513), Mathematical Physics (PH 511)
Intended for	M. Sc (Physics), Ph.D, iPhD, B. Tech 3rd and 4th year
Distribution	Elective for M. Sc (Physics), Ph.D, iPhD, B. Tech 3rd and 4th year
Approval	recommended by the 37th BoA (Oct 2020)

Course Contents

- **Module I**
- Mathematical Preliminaries, Lagrangian and Hamilton density, second quantization, functionals, path integrals, functional field integrals, coherent states for bosons and fermions. [7 Lectures]
- **Module II**
- Classical fields, Klein-Gordon field, massless scalar field theory, massive scalar fields, Phi-4 theory, complex scalar fields. [5 Lectures]
- **Module III**
- Schrodinger, Heisenberg, and interaction pictures, time-evolution operator, translations, and rotations in space-time, transformations of quantum fields, symmetries and conservation laws, Noether's theorem. [8 Lectures]
- **Module IV**
- Canonical quantization of fields with examples, normal ordering, internal symmetries, massive vector fields, polarizations, gauge fields and gauge theory. [7 Lectures]
- **Module V**
- Propagators and Green's function, Dyson equation, field and Feynmann propagator, Smatrix, perturbation expansion, Wick's theorem, Feynmann Diagrams. [7 Lectures]
- **Module VI (Some Application of Field theory in condensed matter physics)**
- Superfields and fields. Fermi liquid theory, field theory formulation of many body problem in metals and superconductors, Hartree-Fock energy, random phase approximation, fractional quantum Hall effect. [8 Lectures]
- or
- **Module VI (Some applications of field theory in particle physics)**
- Dirac and Weyl equation, spinors, transformation of spinors, quantizing the Dirac field, fermion propagator, quantum electrodynamics (QED) scattering cross sections [8 Lectures]

Text Books

1. Lancaster and Blundell, **Quantum field theory for gifted amateur**, Oxford, 2014.
2. Mandl and Shaw, **Quantum field theory**, John Wiley and Sons, 2010.
3. Peskin and Schroeder, **An Introduction to Quantum field theory**, CRC Press, 2018.

Reference Books:

1. S. Weinberg, **Quantum Theory of Fields**, Vol. 1, Cambridge, 1995.
2. M. Srednicki, **Quantum field theory**, Cambridge, 2007.
3. M. H. Ryder, **Quantum field theory**, Cambridge, 1996.

Online Resources

1. David Tong, **Lectures on QFT**: <http://www.damtp.cam.ac.uk/user/tong/qft.html>
2. Freeman Dyson, **Lectures on advanced quantum mechanics**, <https://arxiv.org/pdf/quant-ph/0608140.pdf>